

17.7 Conditional offers

The Bidder should abide by all Terms and Conditions specified in the RFP Document. Conditional offers shall be liable for disqualification.

18. Scope of work

The Directorate of Survey & Land Records, Government of Puducherry intends to engage with competent agency for establishing integrated Web GIS Platform to meet Land Records digitalization, management & dissemination requirements of the State.

The solution shall be aligned with open standards (OGC WMS / WFS / WMTS / WCS), the national ULPIN (Bhu-Aadhaar) framework, the India Digital Ecosystem of Agriculture (IDEA) standards for interoperability with the Farmer Registry and the Digital Crop Survey, and shall consume data from DoSLR's Record of Rights (Nilamagal / equivalent) on a strictly read-only basis. Read-only applies to baseline data pulls; approved mutations initiated through the WebGIS Online Mutation menu shall be written back to Nilamagal through a controlled approval workflow.

18.1 Key Principles of Scope

The Scope of Work and the Agency's execution of the engagement shall be governed by the following principles:

- Parcel-by-parcel FMB reconstruction is the core activity of this engagement.** Every Field Measurement Book (FMB) sheet in the Union Territory shall be reconstructed parcel by parcel from the measurements, bearings, chain lines, offsets and call-outs on the FMB, drawn against the drone orthomosaic supplied by DoSLR. The existing CollabLand / Nilamagal shapefiles and the legacy sheets shall be used as reference inputs only.
- The entire solution shall be built on a fully open-source technology stack.** The WebGIS Platform, the mobile application and all back-end services shall be built on a fully open-source technology stack. All experience that the Bidder claims for

Pre-Qualification and for Technical Evaluation shall also be on an open-source technology stack, supported by work orders and completion certificates.

- The solution shall be hosted on Government-approved cloud infrastructure within India.** The solution shall be hosted on a Government-approved cloud service provider (CSP) that is empanelled by the Ministry of Electronics and Information Technology (MeitY). Hosting on a non-approved CSP or on non-Government on-premise infrastructure shall be treated as non-responsive. Cloud hosting charges shall be reimbursed by DoSLR on an actual-usage pass-through basis at cost, without any mark-up, administrative fee or handling charge. As and when DoSLR notifies readiness of the State Data Centre or State Cloud, the entire workload shall be migrated to it within the period specified by DoSLR. Detailed cloud, disaster recovery, data residency, security and exit requirements are set out in the Cross-cutting Technical Requirements section of this RFP (Sec.18.6).

The scope of this RFP is organised as three (3) Modules as listed below.

- Module 1:** Base Data Preparation & Cadastral Mapping – parcel-by-parcel FMB reconstruction, stitching into village / taluk / region mosaics, drone-orthomosaic georeferencing, rubber sheeting, multi-tier QC, and bi-directional mutation ingest from RoR (Nilamagal application) and vice versa.
- Module 2:** WebGIS Platform Development – data visualisation, core functionalities, editing tools, RoR attribute integration, AgriStack Farmer Registry and Digital Crop Survey integration, ULPIN management, Discrepancy & Anomaly Analytics with multi-tier QC and role-based user access.
- Module 3:** Automated Georeferencing & Field Integration – Android mobile application for field officers (consumption of DGPS / GNSS

Rover points supplied by DoSLR or its survey partner), web-based point ingestion and visual georeferencing workbench inside the WebGIS Platform, and a server-side automated georeferencing engine with multi-tier QC.

The detailed scope of each Module is described in the following sections. Expected deliverables and cross-cutting requirements (scalability, OGC compliance, security, RBAC, timelines, milestones, SLAs and technical architecture) are specified at the end of this Section.

18.2 Module 1: Base Data Preparation & Cadastral Mapping

The objective of Module 1 is to reconstruct the cadastral fabric of the Union Territory parcel by parcel from the FMBs, stitch the reconstructions into seamless village / taluk / region mosaics, and georeference the mosaics to the drone orthomosaic and to the ground control points supplied by DoSLR. This is the core and most technically demanding activity of the engagement. The existing CollabLand / Nilamagal shapefiles and legacy sheets shall be used as reference inputs only. The Bidder shall deliver all work on a fully open-source technology stack.

18.2.1 Source Data Collation, Stocktaking and Reference Inputs

- Collect and inventory the primary reconstruction inputs: (a) the digital FMB sheets supplied by DoSLR, which carry the measurements, bearings, chain lines, offsets and call-outs of each parcel, per village / FMB number; and (b) the drone orthomosaic supplied by DoSLR or its survey partner, which is the reference base map against which every FMB is drawn.
- Collect and inventory the reference inputs, which shall be used as cross-check inputs only: CollabLand / Nilamagaldigitised FMB shapefiles with attribute tables (Survey No., Sub-Div No., extent, classification, owner reference key);vector village maps; digitized RoR extracts; prior survey reports; and historical mutation registers.

- Prepare a Source Data Register per village listing: FMB sheet count and legibility grade; number of parcels expected from the FMB record; number of parcels present in the CollabLand shapefile; scale and year of the source survey; data provenance; and any known gaps or mismatches against the RoR parcel count.
- Establish – in consultation with DoSLR – the reconstruction priority order (village / taluk / region sequence), the parcel-level exception register for illegible / missing FMB sheets, and the escalation protocol for resolving discrepancies between the FMB sketch, the CollabLand vector, the legacy sheets and the RoR record.

18.2.2 FMB Reconstruction (Parcel-by-Parcel, from the FMB)

- Sheet preparation: load the digital FMB sheet supplied by DoSLR into the reconstruction environment and geo-rectify the FMB to its approximate on-ground footprint using the drone orthomosaic and any on-sheet reference ticks, north arrow and scale bar.
- Reconstruction from measurements: for every parcel on every FMB, compute and draw the parcel polygon from the bearings, chain lines, offsets and call-outs written on the FMB; apply the recorded scale and orientation; and honour the on-sheet topology (adjacency, shared sides and interior divisions) exactly as shown by the Department surveyor/ Draughtsman.
- Base map anchoring: while drawing, use the drone orthomosaic as the reference base map so that physically visible features (roads, bunds, compound walls, water bodies, buildings) are to be matched; where the FMB-computed polygon conflicts with the orthomosaic, log the conflict in the reconstruction exception register.
- Cross-checks (not substitutes): compare the reconstructed polygon with (a) the CollabLand vector where it exists; (b) legacy sheets where it exists; and (c) the RoR recorded extent. Deviations beyond a configurable threshold shall be flagged for DoSLR review. The

reconstructed polygon remains the candidate for the new layer unless DoSLR directs otherwise.

- Closure and accuracy targets for individual FMB reconstruction: geometric closure error $\leq 1/1000$ of the perimeter; attribute completeness $\geq 99\%$; parcel area deviation $\leq 3\%$ against the FMB-written extent.
- Exception handling: FMB sheets that are illegible, missing or internally inconsistent shall not be forced through reconstruction. They shall be entered in a written exception register, triaged with DoSLR, and resolved through targeted re-surveying under DoSLR's separate survey programme or through accepted use of the CollabLand vector as a provisional placeholder pending re-survey.
- QC on reconstructed FMBs: every reconstructed FMB shall pass through rule-based automated QC (closure, topology and extent) and visual QC against both the digital FMB and the drone orthomosaic. Minimum 10% cartographer sample per village and 100% of exception-register parcels.

18.2.3 Mosaicing Reconstructed FMBs into Village / Taluk / Region Maps

- Edge-matching: reconcile shared edges between adjacent FMBs by reading the abutment call-outs on each FMB and snapping within a DoSLR-approved tolerance; where the two FMBs disagree on the shared edge, log the conflict and resolve per the reconstruction exception register (not by arbitrary midpoint averaging).
- Village-level stitching: produce a topologically clean village map (no gaps, no overlaps, no slivers, single-ringed exterior geometries, closed polygons); maintain the hierarchical topology (Parcel $\rightarrow$ FMB $\rightarrow$ Village $\rightarrow$ Taluk $\rightarrow$ Region) with a lineage key from every parcel back to its source FMB sheet and page.
- Resolution of conflicts with reference datasets: if the stitched village map conflicts with the CollabLand vector or village map, the

stitched reconstruction prevails; conflicts are logged with before/after snapshots for DoSLR adjudication; the CollabLand vector shall be validated before reconstruction with due approval.

- Taluk / region mosaic: Enroll village maps up to taluk and region level ensuring consistency of common boundaries and administrative overlays; publish a per-village and per-taluk mosaic QC report.

Georeferencing the Reconstructed Mosaic

- Georeference the reconstructed village / taluk / region mosaic to a uniform spatial reference system (as specified by DoSLR) using a combination of (a) drone orthomosaic imagery provided by DoSLR or its survey partner, and (b) Ground Control Points (GCPs) supplied by DoSLR.

Note: All spatial data shall be in WGS 84 (EPSG:4326) for storage, with display in India Zone II (Lambert Conformal Conic / Everest datum) or as directed by DoSLR in the Project Inception Report

- Apply affine / polynomial / spline / thin-plate spline transformation as appropriate to the local distortion; document the transformation choice per village and justify in the QC report.

18.2.4 Quality Control (Cadastral Layer)

- Three-tier QC: (i) rule-based automated QC (topology, attribute, spatial reference); (ii) visual QC by trained cartographers with sampling plan; (iii) field spot-check in coordination with DoSLR for a minimum 1% sample, with results reconciled before acceptance.
- Publish a per-village QC Report documenting error categories, resolutions, residuals and a signed acceptance from DoSLR.
- Minimum acceptance criteria: $\geq 95\%$ parcels pass positional accuracy(Positional accuracy threshold: $RMSE \leq 1.0$ m at cadastral scale, in accordance with DILRMP standards for cadastral mapping); 100% pass topology validation; $\geq 99\%$ attribute completeness.

18.2.5 Bi-directional Mutation Ingest from RoR and back to ROR

- Implement a two-way, mutation-ingest pipeline that consumes RoR mutation events – (i) sub-division, (ii) amalgamation, and (iii) attribute-only changes (owner, tenant, classification) – from DoSLR's Nilamagal / RoR system and applies the corresponding geometry and attribute updates on the cadastral layer. The direction of flow is both way-RoR to WebGIS and vice versa.
- Each ingested mutation shall produce an audit record with before/after geometry, attributes, timestamp, linked RoR transaction identifier and ULPIN lineage. The cadastral layer shall be time-aware: queryable as of any date.

18.3 Module 2: WebGIS Platform Development

The objective of Module 2 is to design, build, host and operate a secure, scalable, OGC-compliant WebGIS Platform that serves the cadastral layer and integrated attribute data to DoSLR officials, allied line departments, and – in a controlled manner – to citizens. The platform shall be accessible through modern web browsers and shall expose open APIs for programmatic consumption.

18.3.1 Data Visualisation

- Drone orthomosaic layer – published as a raster / tiled service (WMTS / XYZ) with base visualisation at 1:500 and progressive zoom to village / taluk / region scale. Visualisation shall support 1:250000 (UT overview) to 1:500 (individual parcel inspection) with scale-dependent rendering at each zoom level.
- Parcel layer (polygon) – rendered with boundary styling, parcel labels (Survey No. / Sub-Div), and symbology driven by attribute classes (e.g., land classification, ownership type).
- FMB layer – the reconstructed FMB sheet outlines and their call-outs, toggleable for verification.

- Administrative boundaries – Region, Taluk, Village, Ward and block boundaries.
- Base maps – OSM, hybrid satellite, and a DoSLR-approved base map. Users shall be able to switch base maps and control layer opacity.
- Rendering performance target: tile response ≤ 300 ms at p95 for pre-cached tiles; map load ≤ 2 s at p95 for typical village views.

18.3.2 Core Functionalities

- Parcel search by Survey Number and Sub-Division Number, by Village + Survey No., by ULPIN, by owner name (where authorised), and by address/location text. Owner name search is restricted to authenticated, authorised officers only. All such queries shall be logged. No owner PII shall be visible in public or anonymous access tiers.
- Navigation and zoom: pan, zoom in/out, zoom to extent, zoom to bookmark, scale bar, coordinate display (lat/long and projected).
- Measurement tools: distance (geodesic and planar), area (hectares / sq. m / sq. ft / acre / cent), bearing, buffering.
- Spatial queries: point-in-parcel; parcels within distance; parcels intersecting a drawn polygon; parcels along a corridor; parcels within an administrative unit.
- Attribute display: on click of a parcel, a side panel shall display the full attribute record (geometry-derived extents, RoR attributes, mutation history, ULPIN, linked FR records for agriculture parcels).
- Print / export: high-resolution PDF map export with legend, scale and title block; CSV / Shapefile / GeoJSON export subject to role permissions. All export events are logged with user, timestamp, and geographic extent.

18.3.3 Editing Tools

- Polygon editing: vertex add / move / delete; parcel sub-division by drawing a split line; parcel amalgamation by selecting adjacent

parcels; snapping and tolerance controls. Edits shall be committed directly by designated power users and shall be captured in an immutable audit log.

- Rubber sheeting (within the tool): designated power users shall be able to apply rubber-sheet transformations to a selected area, using additional control points, with preview and revert.
- Version control and audit: every change shall be versioned; the platform shall support rollback to any previous version; an immutable audit log shall record user, timestamp, action, before/after geometry + attributes.

Note: Version history shall be retained for minimum 7 years on hot/warm storage with instant retrieval. Older versions may be archived to cold storage with retrieval SLA of 24 hours.

18.3.4 Revenue Data Integration

- The WebGIS Platform shall maintain a bidirectional integration with DoSLR's Nilamagal (RoR, Patta, Chitta, Adangal) and Collabland systems, consuming parcel-level data through secure APIs or change-data-capture while facilitating updates via an Online Mutation menu. Any data modifications initiated within the WebGIS environment must undergo formal digital approval by the designated DoSLR official before being synchronized back to Nilamagal and Collabland, ensuring that both textual records and spatial geometries are consistently updated across all three platforms upon authorized validation.
- Attribute fields shall include (illustrative): Survey No., Sub-Div No., Village, Taluk, Region; Owner(s) and share; Tenant(s) and share; Classification (Nanjai / Punjai / Manavari, etc.); Extent as recorded in RoR; Land Revenue / Kist; Encumbrances; Mutation history with dates.

- Support certified extracts: the platform shall generate a parcel extract combining map and RoR attributes, digitally signed, issuable via DigiLocker or other facilities admitted by DoSLR.
- The WebGIS Platform shall incorporate a dedicated Online Mutation menu to facilitate the provision of updated land records for the AgriStack project, adopting an integrated digital workflow similar to the model implemented in neighboring States. This module will enable the end-to-end processing of land record modifications directly within the GIS interface; upon formal digital approval by the designated DoSLR authority, the system shall automatically trigger synchronized updates to both the Nilamagal textual database and Collabland spatial records, ensuring a unified and real-time data flow essential for the national AgriStack ecosystem.

18.3.5 Agriculture Data Integration (AgriStack – Farmer Registry and Digital Crop Survey)

- API integration with the AgriStack Farmer Registry (Farmer ID) maintained by the Department of Agriculture & Farmers Welfare – so that every agriculture parcel on the cadastral layer can be linked to its cultivator(s) / tenant(s), and so that Farmer ID → parcel and parcel → Farmer ID lookups are supported on the WebGIS.
- API integration with the Digital Crop Survey (DCS) / Crop Sown Registry – so that the WebGIS can display the current-season crop sown, historical crop rotation, and crop-season summaries at parcel / village / taluk / region scale.
- Reverse integration: publish authoritative parcel geometry, ULPIN and RoR-linked attributes back to the Farmer Registry and DCS so that the agriculture stack relies on a single authoritative cadastral spine maintained by DoSLR.

- The integration shall be consent-based and conform to the India Digital Ecosystem of Agriculture (IDEA) framework and the AgriStack Reference Architecture; no personally identifiable information (PII) shall be displayed in public / citizen views; use tokenised references where required by UIDAI and DPDP Act norms.

Agency shall submit a Data Protection Impact Assessment (DPIA) per the DPDP Act, 2023 as part of the SRS deliverable, and shall execute a Data Processing Agreement with DoSLR before accessing any personal data of citizens.

18.3.6 Unique Land Parcel ID (ULPIN / Bhu-Aadhaar)

- Generate, assign and reconcile the 14-digit Unique Land Parcel Identification Number (ULPIN, also known as Bhu-Aadhaar) for every parcel in the Union Territory, in line with the national ULPIN specification published by the Department of Land Resources, Ministry of Rural Development.
- Handle ULPIN lifecycle on mutation: sub-divided parcels receive new ULPINs; parent ULPIN is retired with a full lineage record; amalgamated parcels receive a new ULPIN; the parents are retired with lineage.
- Expose ULPIN lookup and resolution APIs; integrate with the national ULPIN service where published; maintain a mapping register to cross-reference any legacy parcel identifiers.

18.3.7 User Access and Roles

- Role-based access control (RBAC) with, at minimum, the following role classes: Public Citizen (anonymous, limited attributes); Authenticated Citizen (DigiLocker / MobileOTP); Village / Taluk Officer; Revenue Inspector; Surveyor; DoSLR Officer; Line Department Officer (Agriculture, PWD, Urban, etc.); Administrator; Auditor.

- Public / citizen view: parcels, boundaries, Survey No., village / taluk / region context, and non-sensitive metadata; no owner PII.
- Officer view: full RoR attributes, mutation history, cadastral editing tools (subject to role).
- Authentication: SSO for officers where authorised; DigiLocker or OTP-based authentication for citizens; MFA for privileged roles.

Note: Agency shall deliver a Role-Based Permission Matrix as part of SRS, specifying create/read/update/delete/export permissions per role. Matrix to be approved by DoSLR before UAT

18.3.8 Discrepancy and Anomaly Analytics (Visual Support for Officials)

The WebGIS Platform shall provide a GIS analytics layer that compares the RoR record with the cadastral map and shows the differences visually for the review of authorised officials. The capability is a visualisation and analytics tool that supports the existing departmental processes of DoSLR.

- Area variance analysis – for every parcel, compare the RoR-recorded extent with the geometry-computed extent; classify the absolute variance into configurable bands (default: Green $\leq 1\%$, Amber 1–5%, Red $> 5\%$) and render the parcel in the corresponding colour on a dedicated "Variance" map layer.
- Boundary mismatch analysis – overlay the cadastral vector on the drone orthomosaic / high-resolution imagery and flag parcels whose boundary deviates from the visible on-ground feature by more than a configurable tolerance; colour-code the deviation and highlight the specific segment(s) of the boundary at issue.
- Record-map mismatch dashboard – consolidated view of: (i) parcels with missing RoR records; (ii) RoR records with missing geometry; (iii) parcels with duplicate / overlapping geometry; (iv) ULPIN conflicts; (v) classification mismatches; (vi) ownership/tenancy conflicts against Farmer Registry.

- Visual measurements on the map – on-map display of computed parcel area, perimeter, neighbouring parcel extents, centroid coordinates and RoR extent in a side panel and as on-map annotations, so that officials can analyse visually without leaving the map.
- Filter and drill-down – officials shall be able to filter anomalies by region / taluk / village / ward / variance band / anomaly type; each row shall open to a pre-zoomed map view with the issue highlighted.
- Exportable reports – anomaly lists as PDF / Excel; variance heat-maps as print-ready cartographic outputs, so that officials can carry them into their existing departmental processes.

18.4 Module 3: Automated Georeferencing & Field Integration

Module 3 comprises (i) a mobile application for field officers, (ii) a browser-based point ingestion and visual georeferencing workbench inside the WebGIS Platform, and (iii) a server-side engine that consumes inputs from both surfaces and refines the cadastral layer. All DGPS / GNSS control points required for this Module shall be supplied by DoSLR or its separately appointed survey partner and shall be consumed by the Agency. The entire Module 3 shall be built on a fully open-source technology stack.

18.4.1 Mobile Application for Field Integration

- Android application (Android 10 or higher), built on a fully open-source technology stack. Target devices are standard Android tablets and phones used by field officers.
- Map viewing: map view of the cadastral layer, drone orthomosaic and administrative boundaries for the officer's assigned village; parcel search by Survey or Sub-Division number; pan, zoom, measure and parcel attribute display (on a read-only basis).
- DGPS / GNSS point consumption: ingestion of DGPS / GNSS control points supplied by DoSLR or its survey partner over

Bluetooth, NTRIP or file import, for the purpose of on-field georeferencing. The points so consumed shall sync to the WebGIS Platform and feed the automated georeferencing engine and the visual georeferencing workbench.

- Security: on-device encryption; role-based authentication of field officers; and time-limited session tokens.

18.4.2 Web-Based Point Ingestion and Visual Georeferencing

Authorised officials shall be able to perform point ingestion and georeferencing from the browser with the following capabilities:

- Excel / CSV upload of control points – a standard template (ID, description, latitude, longitude, easting, northing, datum, accuracy, source, notes) with validation on upload, preview of rejected rows, and ability to bulk-promote accepted rows into the GCP library for a selected village / cluster.
- Manual coordinate entry – a form where an official can type or paste one or more coordinates (decimal degrees or projected) and drop them as GCPs on the map; visual markers update live.
- Ingestion of mobile-app captured points – DGPS / GNSS points consumed through the mobile application shall stream into the same GCP library in near real time and become immediately visible to web users with appropriate role permissions. Officials shall be able to accept, reject or flag each point with a remark.
- On-map placement and visual verification – ability to pick a GCP on the map, snap it to the drone orthomosaic, and see the cadastral vector "rubber-sheet" in a live preview; transformation type (affine / polynomial / spline / thin-plate spline) shall be selectable with immediate visual feedback and RMSE readout.
- Visual georeferencing of selected parcels / clusters – an officer shall be able to select a parcel, a cluster, or an entire village, pair source points with target points (drag-and-link on the map), and

apply a bounded georeferencing locally without destabilising the rest of the mosaic; dry-run preview is mandatory before committing.

- Audit logging – every web-initiated georeferencing commit shall produce a full audit record (user, timestamp, input points, transformation, RMSE, before/after geometry) equivalent to mobile-initiated edits.

18.4.3 Automated Georeferencing Engine

- Server-side georeferencing pipeline that consumes DGPS / GNSS points (from the mobile app, the web workbench or bulk upload) and drone-orthomosaic anchor points, and re-aligns the affected portion of the cadastral layer in a bounded, explainable way.
- The engine shall support: (a) local re-georeferencing of a village or parcel cluster without destabilising the rest of the mosaic; (b) weight-based merging of multiple inputs (DGPS > drone orthomosaic > legacy reference; Default weights: DGPS=1.0, drone orthomosaic=0.7, legacy reference=0.3, adjustable through admin console with DoSLR approval. Weighting methodology documented in Architecture Document.); (c) dry-run mode for QC before committing; (d) explainability – every edit accompanied by a justification log naming the inputs and transformation.
- Engine-produced edits shall be committed directly after dry-run QC by authorised power users and shall be captured in a full, immutable audit log.